



Efficient silicon-erbium photonic hybrids with flexible spatial control of light via bound states in the continuum

PAI ZHOU,¹ HUI-ZHEN ZHANG,¹ YU-HUI CHEN,^{1,2}  AND XIANGDONG ZHANG^{1,3}

¹Key Laboratory of Advanced Optoelectronic Quantum Architecture and Measurements of Ministry of Education, Beijing Key Laboratory of Nanophotonics & Ultrafine Optoelectronic Systems, School of Physics, Beijing Institute of Technology, Beijing 100081, China

²stephen.chen@bit.edu.cn

³zhangxd@bit.edu.cn

Abstract: The hybrid of silicon and erbium-doped crystals provides a promising photonic platform for the advancement of quantum technologies, combining scalability, telecom compatibility, and complex quantum functionalities. However, the inherent high refractive index of silicon leads to optical field confinement within the silicon, limiting the number of optically active erbium ions and the achievable light absorption, which are essential for various quantum applications. To fully utilize the potential of erbium-doped crystals, we implemented bound states in the continuum within the hybrid silicon structures, which achieve substantial optical energy confinement within the lower-index crystal. Our structure, which is applicable to both transverse magnetic and transverse electric polarization modes, effectively increases the number of optically active erbium ions and demonstrates simultaneous minimal radiative losses and enhanced erbium absorption. Moreover, the core design principle is extensible to other low-refractive-index materials.

© 2025 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](#)

1. Introduction

Silicon photonics, with its advantages in miniaturization [1,2], scalability [3–6], and compatibility with existing telecommunication infrastructure [7,8], offers significant advantages for integrated optics. However, for many quantum applications that require efficient light emission and strong light-matter interactions, silicon's intrinsic properties, namely its weak nonlinearity and lack of inherent light emission, pose serious challenges.

To address these limitations, there is growing interest in integrating materials with different optical properties onto silicon platforms for scalable photonic circuits [9–12]. Among these, erbium-doped crystals, particularly promising for advanced quantum applications, have attracted significant attention. This hybrid approach seeks to exploit well-established silicon technology while incorporating the advantageous properties of erbium ions, including their telecom-band absorption [13–15], long spin coherence times suitable for quantum memories [16–23], and their capacity for quantum microwave-to-light conversion [24–28]. A core requirement to realize the potential of erbium is to ensure that the erbium ions introduced on the chip are optically active, specifically capable of absorbing light [23]. However, the refractive index of silicon ($n_{\text{Si}} = 3.4$ at $1.5 \mu\text{m}$) is significantly higher than that of most host crystals, leading to a strong confinement of optical energy within the silicon. This confinement severely restricts the number of optically active erbium ions and directly leads to a greatly reduced light absorption [29–31], posing a significant challenge to achieve the desired quantum functionalities [30–32].

Bound states in the continuum (BICs), first proposed by von Neumann and Wigner in 1929 [33], are rapidly emerging as a powerful platform for integrated photonic circuits. BICs offer several key advantages, such as high quality factors [34–37], strong light confinement [38–40],

and inherent robustness against radiative losses [41,42], making them ideal for applications that require enhanced light-matter interaction [43–45]. Recently, photonic BICs have been demonstrated in low-refractive-index photonic integrated circuits on a high-refractive-index substrate. [36,41]. However, existing approaches often prioritize engineering the structure to reduce propagation losses [41,42,46], while neglecting the potential of directly extracting the light field into the low-index material for enhanced erbium absorption in hybrid silicon photonics.

Here we propose a novel BIC-based waveguide structure specifically designed to overcome the field confinement issue of these hybrid systems. Our structure is composed of a low-refractive-index polymer layer, a thin film of silicon, and a low-refractive-index substrate, the refractive indices at 1.5 μm of which are typically $n_{\text{poly}} = 1.5$, $n_{\text{Si}} = 3.4$, and $n_{\text{sub}} = 1.8$ [47], respectively. Our specific aim is to enhance the penetration of light into the functional material to increase the number of optically active erbium ions, which is essential for many integrated quantum photonic applications. By performing a thorough analysis of BIC conditions for both the transverse magnetic (TM) and transverse electric (TE) polarization modes, we demonstrate that more than 80% of the optical field can be extracted into the low-index region, while simultaneously achieving low propagation losses. The absorption of erbium ions is improved by about a factor on the order of ten compared to conventional hybrid silicon structures. This design employs industrial silicon fabrication processes and is readily extendable to other materials, providing a versatile and scalable platform for producing quantum photonic circuits.

2. Results

The design of our BIC waveguide is shown in Fig. 1(a). It comprises a patterned low-refractive-index polymer layer (blue), a silicon thin film (orange), and a low-refractive-index substrate. As an example, an erbium-doped yttrium orthosilicate (YSO) layer has been used as the substrate in this work. This waveguide geometry is characterized by two distinct regions (Fig. 1(b)). Region I consists of a multilayer structure formed by air, silicon, and the substrate. Region II extends this structure by incorporating an additional 500 nm thick polymer layer on top (subsequent theoretical analysis employs an approximation of an infinitely thick polymer layer, the validity of which will be discussed in a later section). An optical waveguide mode $E(\mathbf{r})\exp(i\beta z - i\omega t)$ propagating along the z -axis with slowly varying envelope approximation can be described by [36]

$$i2\beta \frac{\partial}{\partial z} E(x, z) = [-\partial_x^2 + \beta^2 - n^2(x)k^2]E(x, z) \quad (1)$$

where β is the propagation constant, $n(r)$ is the material refractive index, $k = 2\pi/\lambda$ is the vacuum wavenumber.

The mathematical form of Eq. (1) is analogous to the time-independent Schrödinger equation, where spatial propagation along the z -axis mirrors the role of temporal evolution. By defining an effective potential, $U(r) = \beta^2 - n_{\text{eff}}^2(x)k^2$, we establish a direct connection between photonic modes and quantum particles. This framework provides a valuable framework for analyzing the interaction between guided and continuum modes within the established paradigm of effective potential theory [41].

The multilayer structures in Regions I and II provide confinement in the y direction for the propagating modes of TM and TE. The effective refractive indices n of the corresponding structures were obtained using the finite element method (COMSOL Multiphysics). Figure 1(c) shows the effective refractive index profiles for the lowest order TM mode and the two lowest TE modes, which is then converted to an effective potential profile (Fig. 1(d)) using the inverse relationship described in Eq. (1).

The introduction of the polymer layer, which has a refractive index higher than that of air, leads to an increase in $n(\mathbf{r})$ for both TM and TE modes. Consequently, within the polymer region (Region II), the bound modes are at lower energy levels compared to the surrounding regions.

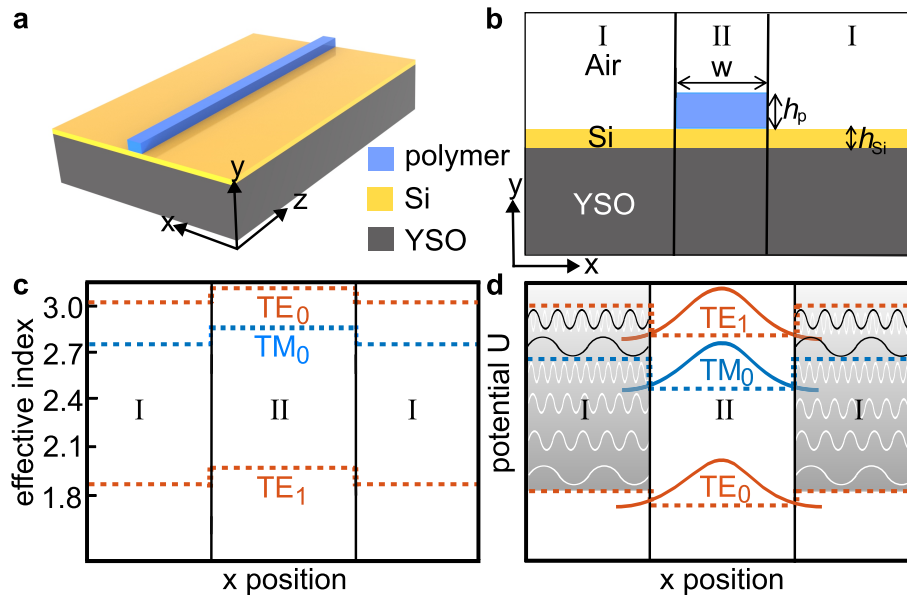


Fig. 1. Illustration of the BIC structure. **a** Illustration of the layered structure, showing the functional substrate (gray, here an erbium-doped YSO crystal), thin silicon film (orange), and patterned polymer layer (blue). **b** Cross-sectional view of the waveguide with key dimensions indicated. **c** Effective refractive index distributions for TE (orange) and TM (blue) polarizations in regions I and II. **d** Corresponding effective potential distributions for TE (orange) and TM (blue) modes. White and black lines indicate the TE and TM continuum modes in region I, respectively.

This results in the formation of potential well structures for both the TE and TM modes, as illustrated in Fig. 1(d). Due to the infinite extent of the silicon layer and crystal substrate in Region I, this structure supports a continuum of propagation modes of both TE and TM. In contrast, the lateral patterning of the polymer on the order of the optical wavelength (Fig. 1(b)) introduces transverse x -direction confinement for both TE and TM modes, leading to discrete bound TE and TM modes within a potential well, which are denoted as TE_0 , TM_0 and TE_1 in Fig. 1(d).

As shown in the potential diagram, the fundamental TE_0 mode within Region II lies outside the range of continuum modes in Region I, resulting in weak coupling between them. However, the bound modes TM_0 and TE_1 in Region II are embedded within, and therefore coexist with, the continuum modes of adjacent Region I. Coupling between these discrete modes (TM_0 and TE_1) and the continuum occurs through scattering at the interface between Regions I and II, causing radiative propagation losses of the guided modes. However, by precisely controlling the width of the polymer layer, it is possible to achieve destructive interference between the scattered waves. This destructive interference, in turn, suppresses the coupling between the bound and continuum modes, ultimately leading to a near-zero radiative loss state, a key feature of an optical BIC [48]. In addition to utilizing the BIC effect in hybrid silicon structures, it is also crucial to ensure that a large portion of the optical energy resides in the functional material. We simulate the optical distribution and propagation loss of our hybrid waveguide structures to validate both of these aspects.

The simulated propagation characteristics of the TE_1 mode are depicted in Fig. 2, showing that typical propagation losses are on the order of 10 dB/cm. A sharp increase to 250 dB/cm is

observed as the polymer width w decreases to $0.5 \mu\text{m}$ (Fig. 2(a)). For $w = 1.2 \mu\text{m}$ (Fig. 2(c)), the field distribution reveals a strong coupling between the guided mode and the continuum modes, explaining the 60 dB/cm loss. In contrast, a notable reduction in propagation loss occurs at specific widths of 0.8, 1.55, 2.3, 3.2, 3.8, and $4.5 \mu\text{m}$, due to destructive interference of scattered waves, as shown in the inset of Fig. 2(a). For the low loss point of $w = 2.3 \mu\text{m}$, the field distribution (Fig. 2(c)) confirms a weak coupling between the guided mode and the continuum modes.

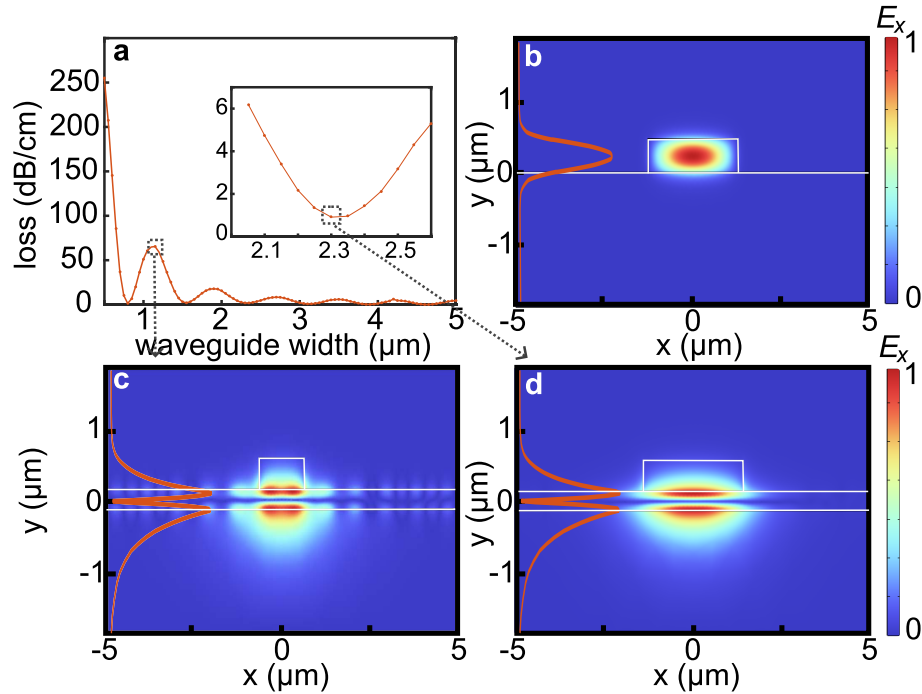


Fig. 2. Simulated results of the BIC waveguide for TE_1 bound mode. **a** Propagation loss as a function of the polymer width w . Inset, Propagation loss of our waveguides at one of the low loss point with polymer width $w = 2.1 \mu\text{m}$. **b** Electric field distribution of a conventional hybrid silicon waveguide, i.e., a silicon waveguide patterned on a YSO substrate. The width of the waveguide is $2.5 \mu\text{m}$. **c** Electric field distribution corresponding to the high-loss point as marked in (a). **d** Electric field distribution corresponding to the low-loss point as marked in inset of (a).

In contrast to traditional silicon waveguides that confine the optical field largely within the silicon material, as shown in Fig. 2(b), the TE_1 mode exhibits a field minimum at the center of the silicon layer (Fig. 2(b) and (c)). This behavior is particularly useful for the integration of functional optical materials into silicon structures, as it enables a large part of the optical field to penetrate the low-refractive-index substrate. This phenomenon arises from the higher-order nature of the TE mode, where the optical field component E_x has opposite signs at the upper and lower boundaries of the silicon layer.

However, a detailed analysis of the low-loss points (Fig. 2(a)) reveals they aren't ideal BICs. For example, the point $w = 2.2 \mu\text{m}$ has a loss of 1 dB/cm (inset of Fig. 2(a)), deviating from the ideal zero loss of BIC. This is because the TE_1 bound mode interacts with both TE and TM continuum modes in Region I. A true perfect BICs (near-zero loss) requires destructive interference of both the TE and TM continuum modes [41], which cannot be guaranteed by

adjusting only the polymer width w . However, the propagation loss of these non-ideal BIC points is already better than commercial amorphous silicon waveguides, which typically exhibits a loss of approximately 6 dB/cm [49]. In our structure, the range of w to achieve a propagation loss below 2 dB/cm is approximately 200 nm.

Unlike the TE_1 bound mode, the TM_0 bound mode exists solely within the TE continuum (Fig. 1(d)), making it possible to achieve near-zero propagation loss under BIC conditions. The simulated performance of the TM_0 bound mode is presented in Fig. 3. With a silicon layer thickness of 350 nm, the typical propagation loss of the TM_0 bound mode is on the order of 1 dB/cm (Fig. 3(a)), significantly lower than the loss associated with the TE_1 bound mode. By varying the waveguide width w , lossless propagation can be achieved at specific BIC points (inset of Fig. 3(g)), which are identified at waveguide widths of 1.1, 2.3, 3.3, 3.8 and 4.4 μm .

In terms of the optical field distribution, the TM_0 bound mode, in distinction to the TE_1 mode, is not subject to a boundary condition that requires the dominant field component to exhibit opposite polarities at the upper and lower boundaries (Fig. 2(c) and (d)). As a result, the TM_0 mode does not necessarily exhibit a field minimum at the center of the silicon layer. Figures 3(b), c, and d, analyzed with a silicon thickness of 350 nm, illustrate this difference. The negligible intensity of the E_x component in Fig. 3(d) confirms that it is a fundamental TM mode. Furthermore, Figs. 3(b) and (c) reveal that the majority of optical energy remains within the silicon layer, with only a small fraction (25%) penetrating into the underlying crystal substrate.

The limited penetration of light is a direct result of the high refractive index contrast between silicon ($n_{\text{Si}} = 3.4$) and the substrate ($n_{\text{sub}} = 1.8$), leading to a strong confinement of the optical field within the silicon layer along the y -direction. To overcome this limitation, we reduced the thickness of the silicon layer to weaken the confinement in the y -direction. Simulations with a silicon thickness of 200 nm (h_{Si}), simulations (Fig. 3(e)-h) still indicate the presence of a TM_0 mode, characterized by the negligible E_x component shown in Fig. 3(h).

In traditional silicon waveguide designs, reducing the silicon thickness leads to a degradation of y -direction confinement and a substantial increase in propagation losses, making them unsuitable for practical light guiding applications. Our structure utilizes a patterned polymer layer ($n_{\text{poly}} = 1.5$) to provide additional confinement along the y -direction. Although this non-ideal confinement still leads to an increase in typical propagation losses (as evidenced by the comparison between Fig. 3(e) and Fig. 3(a)), lossless BIC propagation can still be achieved by adjusting the waveguide width w , with specific BIC points located at $w = 0.85, 1.7, 2.6, 3.4$ and $4.15 \mu\text{m}$ (Fig. 3(e)). A fabrication tolerance of approximately 150 nm is observed to achieve propagation losses below 0.4 dB/cm, as shown in the inset of Fig. 3(e). Because the polymer width could be manufactured with an accuracy of within 10 nm, the BIC effect is easily achievable with existing manufacturing capabilities. Furthermore, the BIC effect also proved robust to variations on the order of 10 nm in polymer and silicon layer thicknesses. In addition, minor deviations in refractive index can be compensated by adjusting geometric parameters like width and thickness, further enhancing fabrication robustness. More importantly, our approach facilitates a significant increase in light penetration into the substrate. As shown in Figs. 3(f) and (g), most of the field is extracted to the polymer layer and the bottom substrate. Approximately 75% of the optical energy is located within the substrate, which leads to better interaction with the functional material.

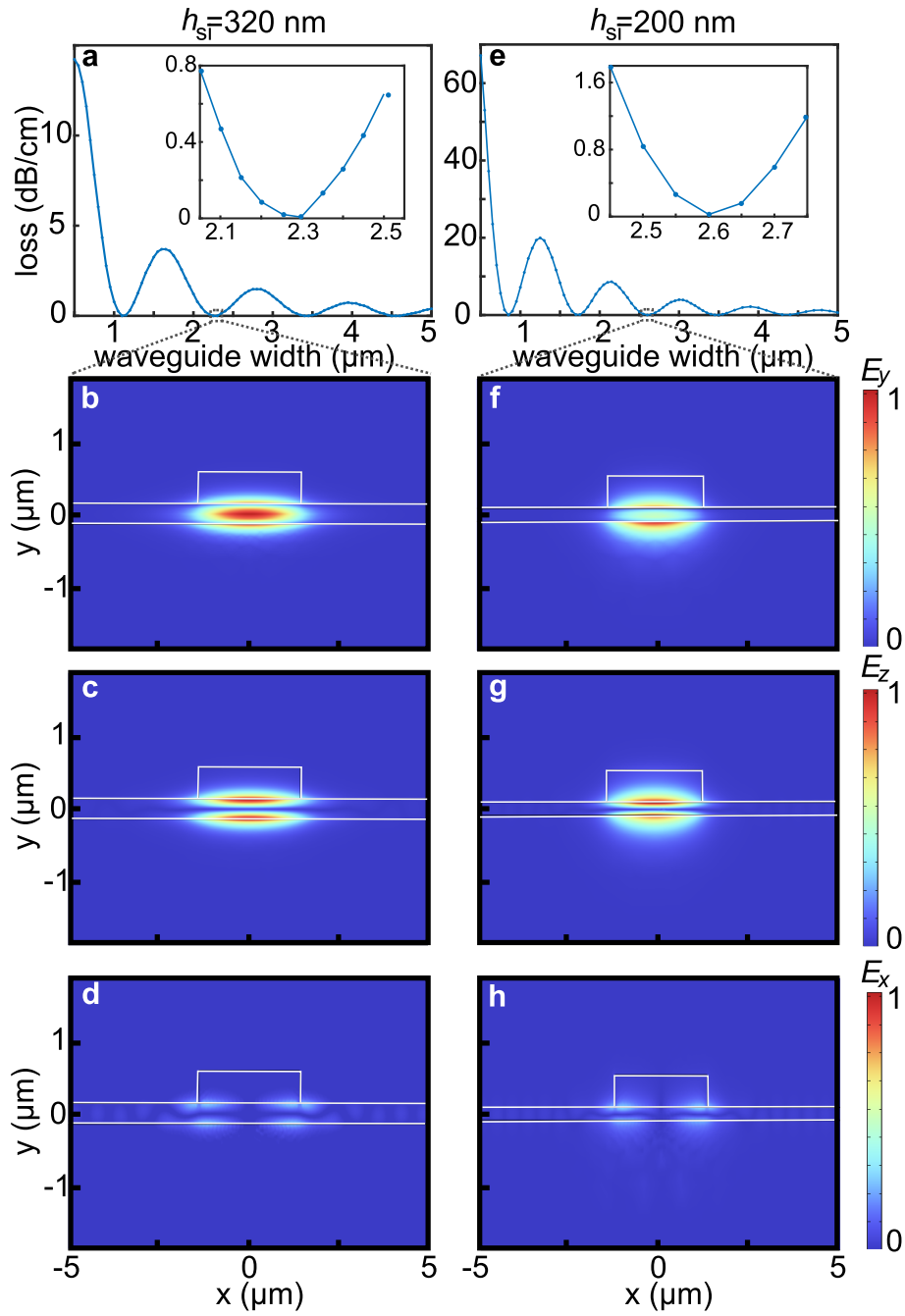


Fig. 3. Simulated results of the BIC waveguide for TM_0 bound mode. **a** Propagation loss of the TM_0 mode as a function of the polymer width w when silicon thickness is $h_{si}=320$ nm. Inset, zoomed-in view of propagation loss $w = 2.3 \mu m$. **b, c** and **d**, optical field distribution of the electric component E_y , E_z and E_x corresponding to the low-loss point at silicon thickness $h_{si}=320$ nm as marked in (a). **e** Propagation loss of the TM_0 mode as a function of the polymer width w when silicon thickness is $h_{si}=200$ nm. Inset, zoomed-in view of propagation loss $w = 2.6 \mu m$. **f, g** and **h**, optical field distribution of the electric component E_y , E_z and E_x corresponding to the low-loss point at silicon thickness $h_{si}=200$ nm as marked in (e).

3. Discussion

Both TE and TM bound modes provide flexible spatial control of light within the low-index substrate. A high rate of field penetration into the substrate indicates a stronger interaction with the embedded erbium ions. We measured the absorption coefficient of a YSO crystal doped with 38 ppm erbium ions and then incorporated this coefficient into the substrate refractive index in our simulations. The simulation results in Fig. 4(a) reveal that for a 500 nm silicon layer, the erbium absorption coefficients for the TE and TM modes are 0.09 cm^{-1} and 0.05 cm^{-1} , respectively. These values are a 22-fold and 13-fold increase compared to a conventional silicon slab on top of an Er:YSO crystal (0.003 cm^{-1}). Reducing the thickness of the silicon layer further enhances the absorption of the TE mode and TM mode to 0.77 cm^{-1} and 1.15 cm^{-1} respectively, as shown in Fig. 4(b). The strong absorption demonstrated here signifies efficient interaction between the light field and erbium ions, which is promising for the development of practical and scalable quantum photonic circuits.

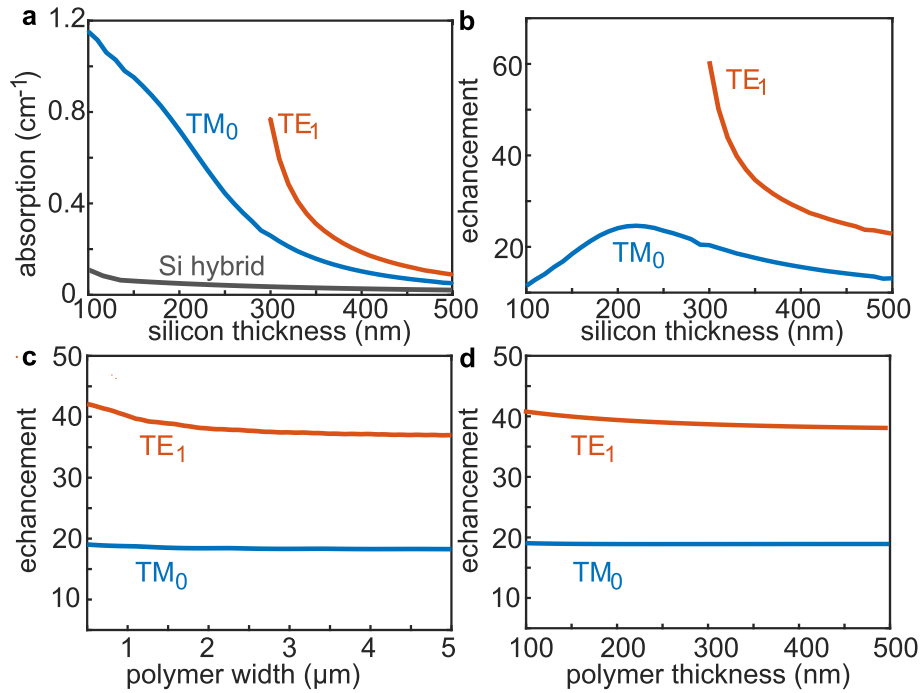


Fig. 4. Enhanced erbium absorption in the BIC hybrid structure. **a** Absorption coefficient of erbium ions in our BIC structure. Blue, TM₀ of the BIC waveguide; orange, TE₁ of the BIC waveguide. For comparison, absorption coefficient of conventional silicon slab on top of a YSO crystal is also shown in gray. The configuration of the conventional hybrid silicon waveguide is as shown in Fig. 2(b). **b** Absorption enhancement of the TM₀ and the TE₁ mode of our BIC structure compared to the conventional hybrid waveguide. **c** Absorption enhancement as a function of polymer width w for $h_{\text{Si}} = 320 \text{ nm}$. **d** Absorption enhancement as a function of polymer thickness h_p for $h_{\text{Si}} = 320 \text{ nm}$. Enhancement factors are compared to the fundamental TE mode in conventional silicon waveguides.

When the silicon thicknesses are larger than 300 nm, our structure supports both TM₀ and TE₁ BIC modes. As silicon thickness decreases, the field penetration of both modes increases due to reduced field confinement along the y -direction. However, the TE₁ mode consistently achieves higher absorption coefficient, reaching up to 60 times higher than the conventional silicon slab

on top of a YSO crystal, compared to only 20 times for the TM mode under same conditions, as shown in Fig. 4(b). Note that these enhancement factors were compared to the fundamental TE mode in conventional silicon waveguide with the same dimensions. This difference is attributed to the inherent property of the TE mode of having a field minimum located in the center of the silicon layer (Fig. 4(a)). Note that in a conventional hybrid waveguide as shown in Fig. 3(b), decreasing the silicon thickness reduces field confinement and increases the absorption coefficient, as depicted by the gray line in Fig. 4(a).

Reducing the silicon thickness below 300 nm further results in two key changes: first, it eliminates the necessary boundary conditions for the TE₁ modes, causing them to vanish, as shown in Fig. 4(a) and (b). Second, it drastically alters the field profile of the TM₀ mode. Above 300 nm, the TM₀ mode is characterized by a field maximum in the middle of the silicon layer (Fig. 3(b)). However, as the thickness decreases and the field is no longer effectively confined, the optical field spreads to adjacent materials (Fig. 3(f)). This transition significantly increases the absorption coefficient of the erbium ions, reaching 1.15cm^{-1} at a thickness of $h_{\text{Si}} = 100\text{ nm}$ (Fig. 4(a)). For a thickness of $h_{\text{Si}} = 220\text{ nm}$ (Fig. 4(b)), a maximum 25-fold absorption enhancement is achieved compared to conventional hybrid waveguide. Note that both the disappearance of the TE₁ mode and the spread of the TM₀ mode are triggered by the same reduction in silicon thickness. Although the erbium absorption coefficient can be further improved by reducing the thickness of the silicon film, this approach has several drawbacks. Specifically, it reduces the optical mode confinement, increases propagation losses, and decreases fabrication tolerances near the BIC resonance. Therefore, an optimum thickness of the silicon film has to be chosen when designing the device.

A critical consideration in our design is the potential for mode coupling between the TM and TE modes, which can increase propagation losses. For silicon thicknesses above 300 nm, the structure supports multiple modes, potentially leading to coupling, especially in the presence of waveguide imperfections. To avoid this, we recommend restricting the silicon thickness. Our design incorporates two strategies to address this. For TM mode operation, we limit the silicon thickness to below 300 nm (Fig. 4(a)), ensuring single mode operation. When operating in TE mode, silicon thicknesses between 400 nm (the effective indices for TM₀ and TE₁ are 2.93 and 2.23, respectively) and 500 nm (the indices for TM₀ and TE₁ are 3.08 and 2.55, respectively) are chosen to maintain a sufficiently large effective index difference, minimizing coupling, and ensuring good mode confinement.

Although the width of the patterned polymer layer significantly affects the propagation loss due to edge scattering, it does not alter the efficiency of light escaping from the silicon layer. Therefore, the absorption of erbium ions for a given silicon thickness remains almost constant with varying polymer width, as shown in Fig. 4(c). Similarly, absorption is only weakly affected by the polymer thickness (Fig. 4(d)). This insensitivity arises from the exponential decay of the optical field within the polymer layer, which also justifies the assumption of an infinitely thick polymer layer in the prior analysis. These findings allow for independent manipulation of waveguide width, to achieve low propagation losses using BIC conditions, and silicon thickness, to control optical field-erbium interaction within the substrate.

4. Conclusion

In conclusion, we have presented a BIC-based approach for controlling light penetration into a low-index substrate to increase the number of optically active erbium ions within a hybrid silicon photonic chip. This design overcomes high-index silicon limitations by enabling both low propagation loss and significant optical absorption enhancement, offering both TE and TM polarization options. While the TE₁ mode does not achieve ideal BIC, it provides significant absorption enhancement (60-fold) compared to conventional silicon waveguides of the same dimensions when the silicon layer is thicker than 300 nm. The TM₀ mode, however, achieves

perfect BIC and demonstrates up to 1.15 cm^{-1} absorption coefficient of erbium ions when the silicon thickness is below 300 nm. Our BIC structure is particularly well-suited for applications in the $1.3 \text{ }\mu\text{m}$ and $1.5 \text{ }\mu\text{m}$ telecommunication bands. The relatively low optical loss of silicon in these regions is a key factor influencing this effectiveness. Furthermore, our structure exhibits favorable fabrication tolerances, and is also applicable to other optical functional materials with low refractive index by slightly adjusting the high-index layer thickness to match their refractive index.

Funding. National Natural Science Foundation of China (12174026, 62105033); Start-up Fund of Beijing Institute of Technology; Science and Technology Innovation Project of Beijing Institute of Technology.

Disclosures. The authors declare no conflict of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

References

1. Z. Zhou, B. Yin, and J. Michel, "On-chip light sources for silicon photonics," *Light: Sci. Appl.* **4**(11), e358 (2015).
2. T. Horikawa, D. Shimura, H. Okayama, *et al.*, "A 300-mm Silicon Photonics Platform for Large-Scale Device Integration," *IEEE J. Sel. Top. Quantum Electron.* **24**(4), 1–15 (2018).
3. X. Qiang, X. Zhou, J. Wang, *et al.*, "Large-scale silicon quantum photonics implementing arbitrary two-qubit processing," *Nat. Photonics* **12**(9), 534–539 (2018).
4. J. C. Matthews, A. Politi, A. Stefanov, *et al.*, "Manipulation of multiphoton entanglement in waveguide quantum circuits," *Nat. Photonics* **3**(6), 346–350 (2009).
5. X. Zhou, D. Yi, D. W. U. Chan, *et al.*, "Silicon photonics for high-speed communications and photonic signal processing," *npj Nanophotonics* **1**(1), 27 (2024).
6. H. Larocque, M. A. Buyukkaya, C. Errando-Herranz, *et al.*, "Tunable quantum emitters on large-scale foundry silicon photonics," *Nat. Commun.* **15**(1), 5781 (2024).
7. Y. Urino, T. Shimizu, M. Okano, *et al.*, "First demonstration of high density optical interconnects integrated with lasers, optical modulators and photodetectors on a single silicon substrate," *Optics InfoBase Conference Papers* **20**, 256–263 (2011).
8. J. F. Tasker, J. Frazer, G. Ferranti, *et al.*, "Silicon photonics interfaced with integrated electronics for 9 GHz measurement of squeezed light," *Nat. Photonics* **15**(1), 11–15 (2021).
9. A. W. Elshaari, W. Pernice, K. Srinivasan, *et al.*, "Hybrid integrated quantum photonic circuits," *Nat. Photonics* **14**(5), 285–298 (2020).
10. W. Luo, L. Cao, Y. Shi, *et al.*, "Recent progress in quantum photonic chips for quantum communication and internet," *Light: Sci. Appl.* **12**(1), 175 (2023).
11. A. Orieux and E. Diamanti, "Recent advances on integrated quantum communications," *J. Opt.* **18**(8), 083002 (2016).
12. P. Androvitsaneas, A. B. Young, J. M. Lennon, *et al.*, "Efficient deterministic giant photon phase shift from a single charged quantum dot," *2017 Conference on Lasers and Electro-Optics, CLEO 2017 - Proceedings 2017-Janua*, 1–2 (2017).
13. C. W. Thiel, T. Bttger, and R. L. Cone, "Rare-earth-doped materials for applications in quantum information storage and signal processing," *J. Lumin.* **131**(3), 353–361 (2011).
14. Y. H. Chen, X. Fernandez-Gonzalvo, S. P. Horvath, *et al.*, "Hyperfine interactions of Er^{3+} ions in Y_2SiO_5 : Electron paramagnetic resonance in a tunable microwave cavity," *Phys. Rev. B* **97**(2), 024419 (2018).
15. S. J. Wang, Y. H. Chen, J. J. Longdell, *et al.*, "Hyperfine states of erbium doped yttrium orthosilicate for long-coherence-time quantum memories," *J. Lumin.* **262**, 119935 (2023).
16. M. Rancic, M. P. Hedges, R. L. Ahlefeldt, *et al.*, "Coherence time of over a second in a telecom-compatible quantum memory storage material," *Nat. Phys.* **14**(1), 50–54 (2018).
17. M. H. Jiang, W. Xue, Q. He, *et al.*, "Quantum storage of entangled photons at telecom wavelengths in a crystal," *Nat. Commun.* **14**(1), 6995 (2023).
18. B. Merkel, A. Ulanowski, and A. Reiserer, "Coherent and Purcell-Enhanced Emission from Erbium Dopants in a Cryogenic High-Q Resonator," *Phys. Rev. X* **10**(4), 041025 (2020).
19. D. C. Liu, P. Y. Li, T. X. Zhu, *et al.*, "On-Demand Storage of Photonic Qubits at Telecom Wavelengths," *Phys. Rev. Lett.* **129**(21), 210501 (2022).
20. L. Yang, S. Wang, M. Shen, *et al.*, "Controlling single rare earth ion emission in an electro-optical nanocavity," *Nat. Commun.* **14**(1), 1718 (2023).
21. S. H. Wei, B. Jing, X. Y. Zhang, *et al.*, "Quantum storage of 1650 modes of single photons at telecom wavelength," *npj Quantum Inf.* **10**(1), 19 (2024).
22. X. X. Li, J. X. Wang, Y. H. Chen, *et al.*, "Topologically protected spatial-phase mismatching for cavity-enhanced quantum memories," *Phys. Rev. A* **109**(4), 043511 (2024).
23. X. X. Li, P. Zhou, Y. H. Chen, *et al.*, "Integrated optical quantum memory controlled by the electro-optic effect," *Phys. Rev. A* **108**(1), 012614 (2023).

24. A. Gritsch, L. Weiss, J. Früh, *et al.*, “Narrow Optical Transitions in Erbium-Implanted Silicon Waveguides,” *Phys. Rev. X* **12**(4), 041009 (2022).
25. J. Rochman, T. Xie, J. G. Bartholomew, *et al.*, “Microwave-to-optical transduction with erbium ions coupled to planar photonic and superconducting resonators,” *Nat. Commun.* **14**(1), 1153 (2023).
26. X. Fernandez-Gonzalvo, Y. H. Chen, C. Yin, *et al.*, “Coherent frequency up-conversion of microwaves to the optical telecommunications band in an Er:YSO crystal,” *Phys. Rev. A - At. Mol. Opt. Phys.* **92**(6), 062313 (2015).
27. X. Fernandez-Gonzalvo, S. P. Horvath, Y. H. Chen, *et al.*, “Cavity-enhanced Raman heterodyne spectroscopy in Er^{3+} : Y_2SiO_5 for microwave to optical signal conversion,” *Phys. Rev. A* **100**(3), 033807 (2019).
28. L. A. Williamson, Y. H. Chen, and J. J. Longdell, “Magneto-optic modulator with unit quantum efficiency,” *Phys. Rev. Lett.* **113**(20), 203601 (2014).
29. T. Zhong, J. M. Kindem, J. G. Bartholomew, *et al.*, “Controlled Retrieval,” *Science* **357**(6358), 1392–1395 (2017).
30. A. M. Dibos, M. Raha, C. M. Phenicie, *et al.*, “Atomic Source of Single Photons in the Telecom Band,” *Phys. Rev. Lett.* **120**(24), 243601 (2018).
31. I. Craiciu, M. Lei, J. Rochman, *et al.*, “Multifunctional on-chip storage at telecommunication wavelength for quantum networks,” *Optica* **8**(1), 114 (2021).
32. E. Miyazono, I. Craiciu, A. Arbabi, *et al.*, “Coupling erbium dopants in yttrium orthosilicate to silicon photonic resonators and waveguides,” *Opt. Express* **25**(3), 2863 (2017).
33. J. von Neumann and E. P. Wigner, “On some peculiar discrete eigenvalues,” *Phys. Z.* **30**, 465–467 (1929).
34. Y. Yu, A. Sakanas, A. R. Zali, *et al.*, “Ultra-coherent Fano laser based on a bound state in the continuum,” *Nat. Photonics* **15**(10), 758–764 (2021).
35. S. I. Azzam, V. M. Shalaev, A. Boltasseva, *et al.*, “Formation of Bound States in the Continuum in Hybrid Plasmonic-Photonic Systems,” *Phys. Rev. Lett.* **121**(25), 253901 (2018).
36. C. L. Zou, J. M. Cui, F. W. Sun, *et al.*, “Guiding light through optical bound states in the continuum for ultrahigh-Q microresonators,” *Laser Photonics Rev.* **9**(1), 114–119 (2015).
37. M. V. Rybin, K. L. Koshelev, Z. F. Sadrieva, *et al.*, “High- Q Supercavity Modes in Subwavelength Dielectric Resonators,” *Phys. Rev. Lett.* **119**(24), 243901 (2017).
38. Y. Plotnik, O. Peleg, F. Dreisow, *et al.*, “Experimental observation of optical bound states in the continuum,” *Phys. Rev. Lett.* **107**(18), 183901 (2011).
39. C. W. Hsu, B. Zhen, J. Lee, *et al.*, “Observation of trapped light within the radiation continuum,” *Nature* **499**(7457), 188–191 (2013).
40. Y. Ruan, Y. Li, Z. Hu, *et al.*, “Strong resonance response with ultrahigh quality factor in grating-multilayer systems based on quasi-bound states in the continuum,” *Sci. Rep.* **12**(1), 21471 (2022).
41. X. Sun, “Photonic integrated circuits with bound states in the continuum: Principle and applications,” *Opt. InfoBase Conf. Pap.* **6**(10), 1342–1348 (2019).
42. M. Ahumada, P. A. Orellana, and J. C. Retamal, “Bound states in the continuum in whispering gallery resonators,” *Phys. Rev. A* **98**(2), 023827 (2018).
43. T. Wang, Z. Li, and X. Zhang, “Improved generation of correlated photon pairs from monolayer WS_2 based on bound states in the continuum,” *Photonics Res.* **7**(3), 341 (2019).
44. T. Wang and X. Zhang, “Improved third-order nonlinear effect in graphene based on bound states in the continuum,” *Photonics Res.* **5**(6), 629 (2017).
45. M. Zhang and X. Zhang, “Ultrasensitive optical absorption in graphene based on bound states in the continuum,” *Sci. Rep.* **5**, 8266 (2015).
46. G. Y. Chen, Z. X. Li, Y. H. Chen, *et al.*, “Highly efficient polarization-entangled photon-pair generation in lithium niobate waveguides based on bound states in continuum,” *Opt. Express* **29**(8), 12110 (2021).
47. E. T. Miyazono, “Nanophotonic Resonators for Optical Quantum Memories Based on Rare-Earth-Doped Materials Thesis,” Thesis p. 130 (2017).
48. C. W. Hsu, B. Zhen, A. D. Stone, *et al.*, “Bound states in the continuum,” *Nat. Rev. Mater.* **1**(9), 16048 (2016).
49. Q. Fang, J. F. Song, S. H. Tao, *et al.*, “Low loss ($\sim 6.45\text{dB/cm}$) sub-micron polycrystalline silicon waveguide integrated with efficient SiON waveguide coupler,” *Opt. Express* **16**(9), 6425 (2008).